Assessment of Rogelio Palomo's v2 CSA

NAUSICA calculation record

October 27, 2026

The v2 circuit reduces output noise but also reduces the voltage produced by an arriving ion. Under the paper's input and readout conditions, its input-equivalent noise is 179.1 electrons RMS, compared with 112.7 for the circuit currently used in the paper. The calculation retains the supplied v2 transistor dimensions and gate voltages.

Circuits and conditions

The v2 M4 has $W = 0.54\,\mu$m, $L = 2.4\,\mu$m and gate voltage 0.64 V. Its M3 has $W = 50\,\mu$m, $L = 0.5\,\mu$m and gate voltage 0.6 V. The paper circuit uses M4 with $W = 2.88\,\mu$m, $L = 1.44\,\mu$m and gate voltage 0.556803 V; M3 has $W = 200\,\mu$m, $L = 2\,\mu$m and gate voltage 0.538477 V. All cases use a 3 fF feedback capacitor and 1.8 V supply.

The native v2 conditions are 0.25 pF external input capacitance and 27 °C. The matched comparison uses 1.25 pF and 20 °C for both circuits. A third calculation combines the v2 M4 and its gate bias with the paper's enlarged M3 and its adjusted gate bias.

Reproduction and model syntax

LTspice 17.2.4 for macOS reproduces the supplied Windows 26.0.1 spectrum over 1 Hz–1 GHz with maximum relative difference $7.6e-08$, using the supplied model card unchanged. The card contains `NOIBE` where `NOIB` is intended. The supplied log reports an unknown parameter. The corrected calculation replaces this spelling and closes the model parentheses, using the same corrected card as the paper. At 1 Hz the output noise amplitude density changes from 214.35 to 215.66 $\mu\text{V}/\sqrt{\text{Hz}}$.

Operating point and voltage gain

Circuit	Baseline (V)	$\lvert V_{\text{out}}/V_{\text{in}}\rvert$ at 1 Hz	M4 $1/g_{ds}$ (MΩ)
Paper circuit	0.879	816.6	28.8
v2, native conditions	1.663	13.4	66.7
v2, matched conditions	1.646	16.8	59.9
v2, retuned gate	0.879	941.7	50.3

The voltage ratio is measured by an AC current excitation with the feedback network connected, preserving the operating point. The resistance is M4's intrinsic small-signal drain resistance, read from the operating-point log at its printed precision. It is distinct from the output resistance of the complete amplifier. The matched v2 output lies near the positive supply; M5 has only 30.6 μV between drain and source. The increased M4 resistance therefore accompanies a much lower amplifier voltage gain at the supplied bias.

Changing only the v2 M4 gate voltage to 0.651398349 V restores an output baseline of 0.878932 V. The transistor dimensions, M3 and other biases are unchanged. The per-ion gain then rises to 29.095 μV/ion, demonstrating that the small signal at the supplied bias comes from the operating point. The ENC is 179.7 electrons RMS: output noise rises along with signal gain. This variant is a separate bias-adjustment study.

Ion current, readout and sensitivity

The current is that of a central square pad of 0.5 mm side in an infinite grounded plane, with homogeneous gas in the upper half-space. The weighting potential is one on the pad, zero on the surrounding plane, and zero at infinity. Ions approach centrally at 3 m/s through the final 5 mm, with mean collection times uniformly distributed over 20 ms. Each ion contributes $e\,\Delta\phi_w$ on this finite path. The input waveform is reused from the saved Julia calculation without renormalization or an extra collection pulse.

The voltage is averaged over centred 300 μs windows at consecutive frame boundaries. The measured signal is $D = -(\overline{V}_1 - \alpha\overline{V}_0)$, where $\alpha = \exp(-20\,\mathrm{ms}/\tau)$. Each circuit's τ is obtained from its 100-ion, current-free tail over 21–120 ms. Noise uses the quiescent SPICE one-sided output spectrum over 0.1 Hz–1 MHz, filtered by the same averaging and frame difference. The one-ion gain g defines ENC $= \sigma_D/g$. The final column is checked using finite-count transients on both sides of $S/N = 5$.

Circuit	g (μV/ion)	σ_D (mV)	ENC (e RMS)	Minimum ions
Paper circuit	27.119	3.056	112.7	564
v2, native conditions	1.547	0.185	119.9	597
v2, matched conditions	1.129	0.202	179.1	890
v2 with enlarged M3	1.170	0.166	142.0	707
v2, retuned gate	29.095	5.228	179.7	900

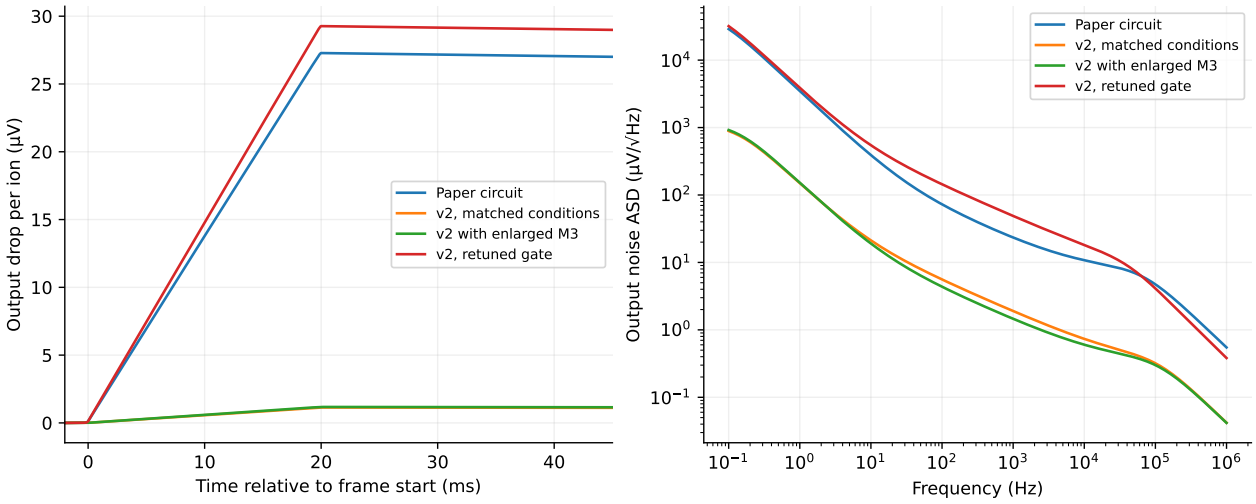


Figure 1: Matched input capacitance and temperature. Left: voltage response to the uniform-arrival current, normalized per ion. Right: quiescent output noise spectrum.

Numerical checks

Both 5 and 2.5 μs maximum time steps are evaluated, with transient tolerances tightened from $(10^{-5}, 10^{-17}\,\mathrm{A}, 10^{-8}\,\mathrm{V})$ to $(5\times10^{-6}, 5\times10^{-18}\,\mathrm{A}, 5\times10^{-9}\,\mathrm{V})$ for relative, current and voltage tolerances. The more stringent inherited voltage tolerance of 10^{-10} V caused the v2 zero-input transient to stall in this simulator. The table gives the successful paired-refinement errors. Frequency densities of 200 and 400 points per decade and independent Gauss quadrature orders of 8 and 16 check the noise integral. Charge conservation is checked against the finite-path input charge.

Circuit	τ (s)	Signal error	Noise error	Fit residual
v2, native conditions	0.6822	7.3e-06	7e-07	1.4e-06
v2, matched conditions	1.2300	7.8e-06	5.6e-07	3.4e-07
v2 with enlarged M3	1.2395	1.2e-05	7.1e-07	3.7e-07

The gain is also checked for 10, 100, 564, 1000 and 2000 ions, and for the counts that bracket the threshold. In the matched v2 case its largest change relative to the one-ion gain is 1.44% over the tested range. This weak nonlinearity is included in the threshold calculation.

Reproduction

Configuration: `config/rogelio_v2.toml`. Acquisition: `scripts/run_rogelio_v2.py`. Julia analysis: `src/ion_signal/rogelio_v2.jl`. Presentation: `scripts/present_rogelio_v2.py`. The pipeline command and full data locations are in `results/ltspice/rogelio_v2/README.md`. Supplied files and the paper's historical results are preserved. This report is generated from HDF5.